4.2.4 Flower symmetry

What is the radius of a circle? Cut a paper into round shape, fold it so as to get two equal halves. In how many planes will you get equal halves? In how many planes you can divide a cucumber in two equal halves? A flower is symmetrical when it is divided into equal halves in any plane running through the center. Flower symmetry is an important structural adaptation related to pollination systems.

1. **Actinomorphic (or) radial or polysymmetric**: The flower shows two mirror images when cut in any plane or radius through the centre.Normally there are more than two planes of symmetry. Example: *Hibiscus, Datura*.



Figure 4.12: (a) Actinomorphic

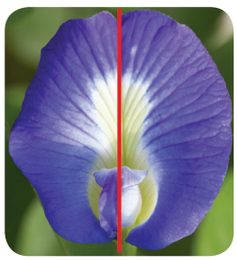


Figure 4.12: (b) Zygomorphic



Figure 4.12: (c) Asymmetric

2. Zygomorphic (bilateral symmetry) or **monosymmetric**: The flower can be divided into 2 equal halves in only one plane. Zygomorphic flower can efficiently transfer pollen grains to visiting pollinators. Example: *Pisum*, Bean.

3. Asymmetric (amorphic): Flower lacks any plane of symmetry and cannot be divided into equal halves in any plane. Parts of such flowers are twisted. Example: *Canna indica*.

4.3 Accessory organs

4.3.1 Arrangement of whorls

The position of perianth (sepals, petals, tepals) parts relative to one another is called **perianth arrangement**.

1. **Cyclic or whorled:** All the floral parts are arranged in definite whorls. Example: *Brassica*.

2. **Acyclic or spiral:** The floral parts are arranged in spirals on the elongated fleshy torus. Example: *Magnolia*.

3. **Spirocyclic or hemicyclic**: Some parts are in whorls and others parts are in spirals. Example:*Annona, Polyalthia*

4.3.2 Calyx

Calyx protects the flower in bud stage. Outermost whorl of flower is calyx. Unit of calyx is sepal. Normally green in colour.

1. Fusion: **a**. **Aposepalous** (polysepalous): The flower with distinct sepals. Example: *Brassica, Annona.*

b. Synsepalous: The flower with united or fused sepals. Example: *Hibiscus*.

2. Duration of floral parts:

What is the green part of brinjal fruit? Have you seen similar to this in any other fruits?

a. Caducous or fugacious calyx: Calyx that withers or falls off during the early development stage of flower. Example: *Papaver*.



Figure 4.13: (a) Caducous bud with sepal



Figure 4.13: (b) Caducous flower without sepal

b. Deciduous: Calyx that falls soon after the opening of flower (anthesis) Example: *Nelumbo*.

11th Botany_EM_Chapter 4.indd 85 05-01-2023 18:22:47